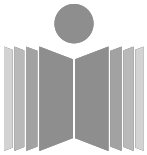
Radioactivity

Saurabh Sandilya

saurabh@phy.iith.ac.in



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

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Radioactivity

- Radioactivity is the spontaneous emission of particles/EM radiation from unstable atomic nuclei.
- This process, also known as radioactive decay, transforms an unstable parent nucleus into a more stable daughter nucleus.
- The emitted radiation can be in the form of particles (like alpha or beta particles) or electromagnetic energy (like gamma rays).
- Radioactive decay is a random process governed by the laws of probability.
- The rate of decay is unique to each radioactive isotope and is measured by its half-life.

Radioactivity: The Accidental Discovery

- 1883: Henri Becquerel started studying fluorescence and phosphorescence.
- 1896: Becquerel started investigating the connection between phosphorescence and X-rays discovered in the same year by Roentgen.
- He placed a uranium salt on a photographic plate wrapped in black paper, hoping sunlight would induce X-rays.
- Due to cloudy weather, the setup was stored in a dark drawer.
- Upon developing the plate, he found the uranium crystal's silhouette clearly exposed the film.



Source: https://www.aps.org/apsnews/2008/02/becquerel-discovers-radioactivity

Radioactivity: The Accidental Discovery

- Becquerel concluded: The uranium salt emitted radiation (which he called 'uranic rays') independently of sunlight or phosphorescence.

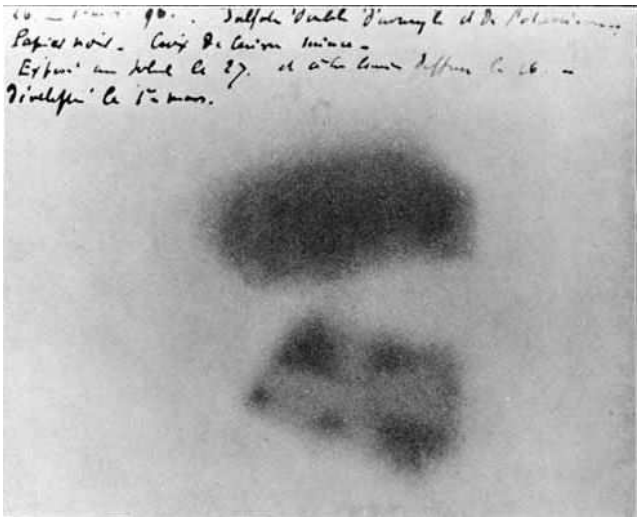
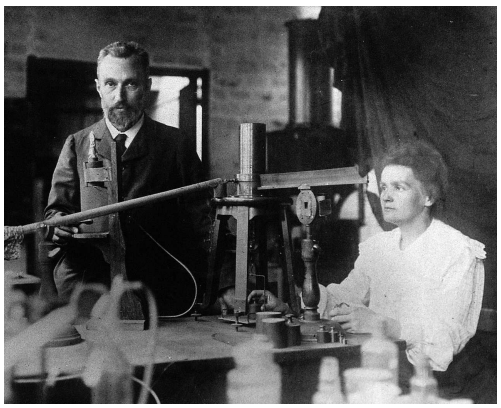


Image of Becquerel's photographic plate, which has been fogged by exposure to radiation from a uranium salt. The shadow of a metal Maltese Cross placed between the plate and the uranium salt is clearly visible.

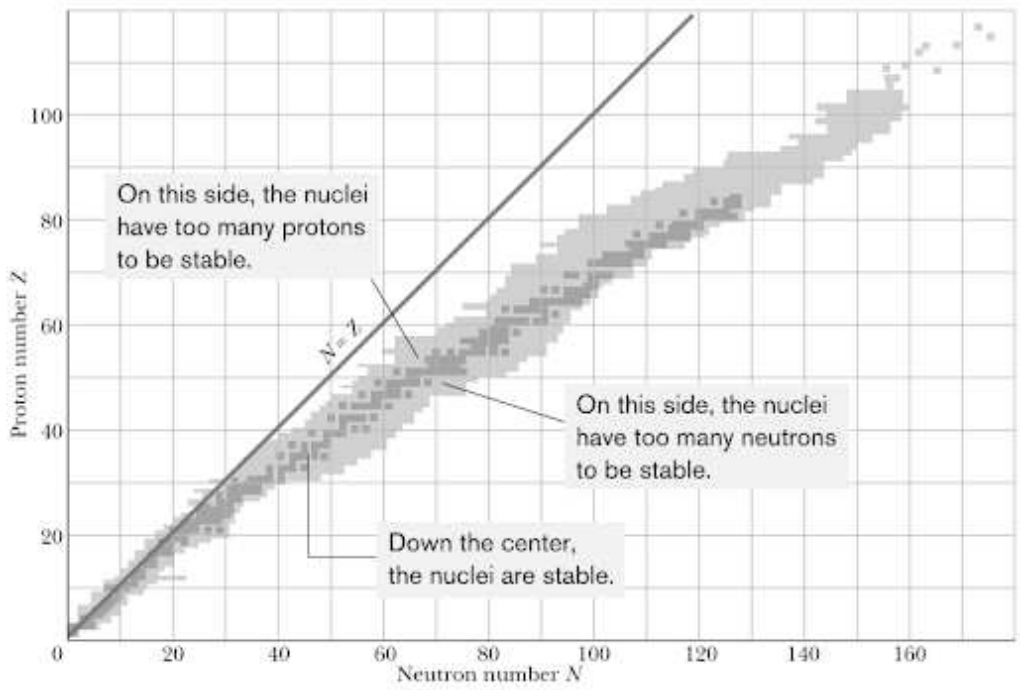
Source: Wikipedia

Radioactivity: The Curies' groundbreaking research



- Madam Curie decided to do her thesis on radiation, recently discovered in uranium by Becquerel.
- Used a sensitive electrometer to measure radiation, establishing that its intensity was proportional to the amount of uranium.
- Coined the term 'radioactivity' to describe this spontaneous emission of radiation.
- Observed that the ore pitchblende was far more radioactive than pure uranium, correctly deducing the presence of undiscovered radioactive elements.
- Through painstaking chemical work with Pierre, she isolated two new, highly radioactive elements: Polonium (named after Poland) and Radium.

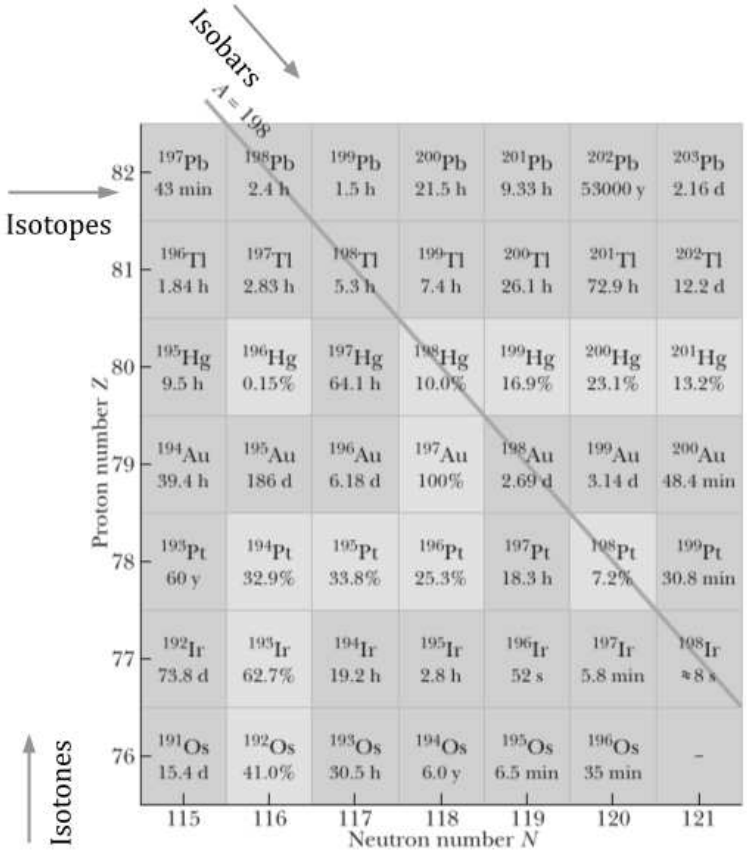
N *vs.* Z plot and nuclear stability



Reference: Halliday/Resnick/Walker

- Low-mass stable nuclides typically possess an equal number of neutrons and protons.
- More massive nuclides exhibit an increasing surplus of neutrons.

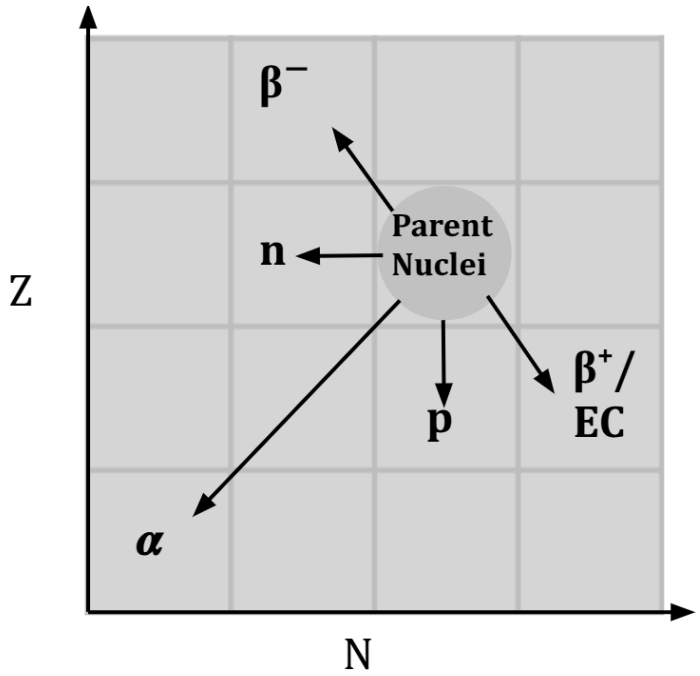
N *vs.* Z zoomed plot



- Isotopes: Same atomic number (Z); different mass numbers (A); and different numbers of neutrons (N).
- Isotones: Different atomic number (Z); different mass numbers (A); and same numbers of neutrons (N).
- Isobars: Different atomic number (Z); same mass numbers (A); and different numbers of neutrons (N).

Radioactive displacement law

- The Radioactive Displacement Law, also known as the Soddy-Fajans-Russell Law, describes how a parent nuclide changes into a daughter nuclide during radioactive decay.



Radioactive displacement law

- **Alpha** (α) **Decay:** A of the daughter nucleus reduces by 4, and Z reduces by 2 (N also reduces by 2).

$$U_{92}^{238} \rightarrow Th_{90}^{234} + He_{2}^{4}$$

- **Beta** (β^-) **Decay:** Z increases by one (N decreases by one), A remains unchanged.

$$n \rightarrow p + e^- + \bar{\nu}_e$$

$$C_{6}^{14} \rightarrow N_{7}^{14} + e^- + \bar{\nu}_e$$

- **Beta** (β^+) **Decay & Electron Capture:** Z decreases by one (N increases by one), A remains unchanged.

(β^+ decay)

$$p \rightarrow n + e^+ + \nu_e$$

$$F_{9}^{18} \rightarrow O_{8}^{18} + e^+ + \nu_e$$

(electron capture)

$$p + e^- \rightarrow n + \nu_e$$

$$K_{19}^{40} + e^- \rightarrow Ar_{18}^{40} + \nu_e$$

The Radioactivity decay law

- Lifetimes of unstable nuclei are statistical in nature.
- If we observe a collection of some unstable nuclei, they don't decay at the same time.
- Consider the disintegration of a 'nuclei $P \rightarrow$ nuclie Q':
  - number of nuclei P decreases with time.
  - number of nuclei Q decreases with time.
- Let N be the (large) number of nuclei P in a sample.
- It is found experimentally that the sample activity (= rate of change of 'N' with time = $\frac{dN}{dt}$) is proportional to the number of nuclei (N) in the sample.

$$\frac{dN}{dt} \propto N$$

$$-\frac{dN}{dt} = \lambda N$$

- λN = Activity of the sample.
- λ = Disintegration constant.

- This means the more nuclei you have, the more decays occur per second.

The Radioactivity decay law

In a sample of large N nuclei, the mean number of nuclei decaying in a time interval dt would be

$$\Rightarrow dN = -\lambda N dt$$

Integrating the above equation

$$\Rightarrow \int_{N(0)}^{N(t)} \frac{dN}{N} = -\int_0^t \lambda dt$$

$$\Rightarrow [\ln N]_{N(0)}^{N(t)} = -[\lambda t]_0^t$$

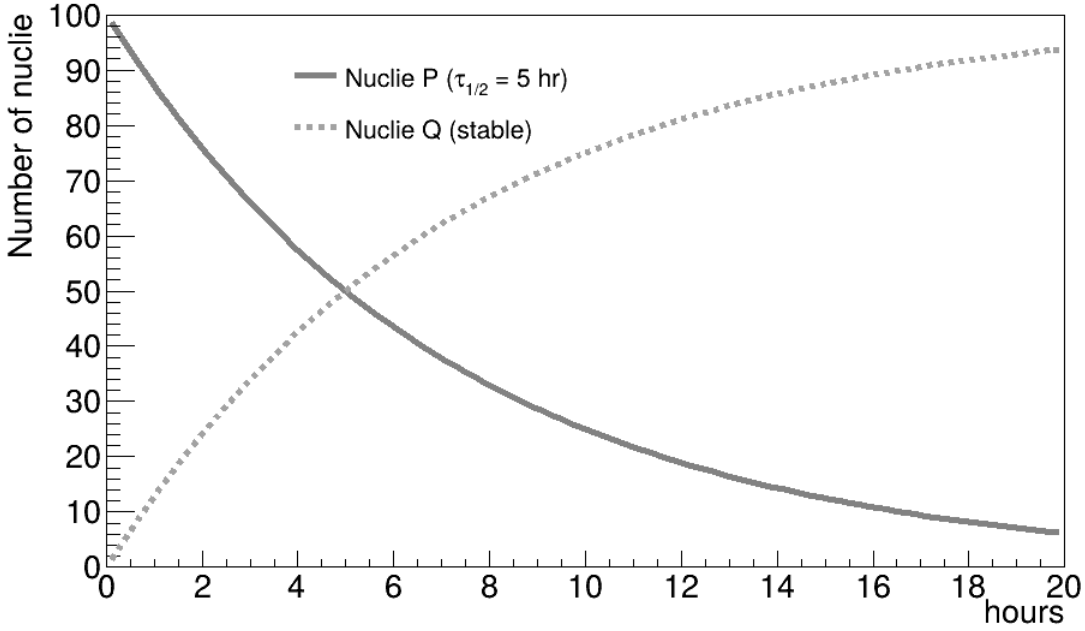
$$\Rightarrow \ln N(t) - \ln N(0) = -\lambda t$$

$$\Rightarrow \ln\left[\frac{N(t)}{N(0)}\right] = -\lambda t$$

$$\Rightarrow \boldsymbol{N(t) = N(0).exp(-\lambda t)}$$

The Radioactivity decay law

- Number of parent (decaying) nuclide (P) sample at t: $N(t) = N(0).exp(-\lambda t)$.
- Number of daughter nuclide (Q) sample at t: $N(t) = N(0).[1 - exp(-\lambda t)]$.



Probability Density for the decay

- Number of nuclei (P) in time $(t + dt)$:

$$N(t + dt) = N(0) \exp[-\lambda(t + dt)]$$
$$N(t + dt) = N(0).\exp(-\lambda t).\exp(-\lambda dt)$$
$$N(t + dt) \simeq N(0).\exp(-\lambda t).(1 - \lambda dt)$$

- So, the number of nuclei decayed between times (t) and $(t + dt)$:

$$N(t) - N(t + dt) = N(0).\exp(-\lambda t) - N(0).\exp(-\lambda t).(1 - \lambda dt)$$
$$N(t) - N(t + dt) = N(0).\exp(-\lambda t).\lambda dt$$

- Probability that an individual nuclie (selected at random) from the initial (at $t = 0$) sample will decay between t and $t + dt$:

$$P(t)dt = \frac{N(0).\exp(-\lambda t).\lambda dt}{N(0)}$$
$$P(t)dt = \exp(-\lambda t).\lambda dt$$

- $P(t)$: Probability that an individual nuclie (selected at random) from the initial (at $t = 0$) sample will decay at time t.

Average lifetime

- To calculate the average/mean lifetime :

$$\tau = \langle t \rangle = \int_0^\infty t.P(t)dt$$
$$\tau = \int_0^\infty t.exp(-\lambda t)\lambda dt = \lambda \int_0^\infty t.exp(-\lambda t)dt$$

- (integration by parts)

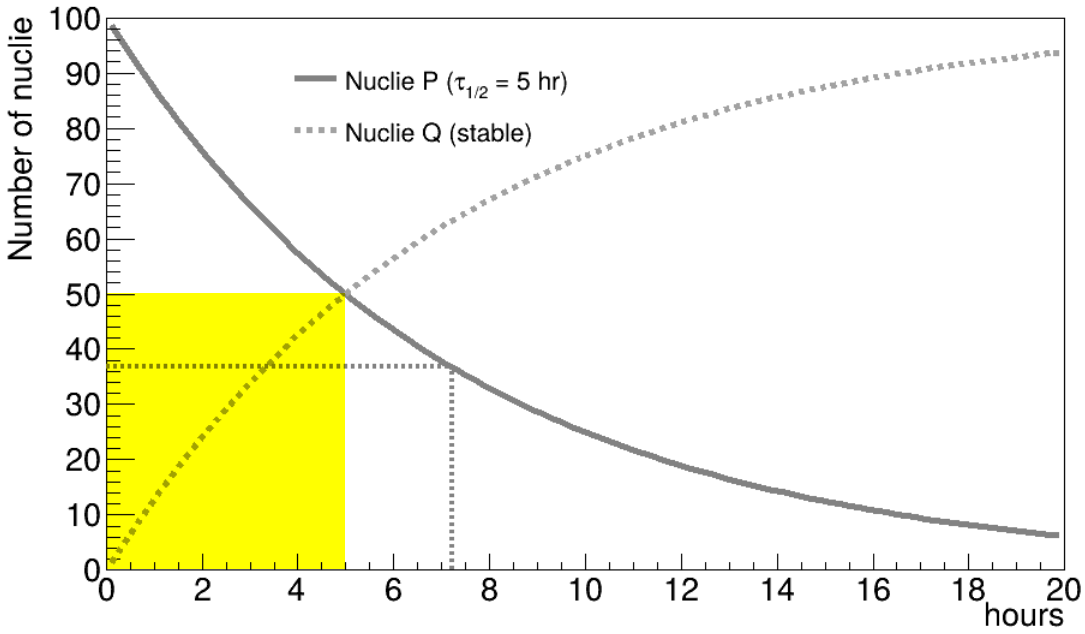
$$\tau = \cancel{\left[-t.e^{-\lambda t}\right]_0^\infty} \to 0 + \int_0^\infty e^{(-\lambda t)}dt$$
$$\tau = \int_0^\infty e^{(-\lambda t)}dt = \left[-\frac{1}{\lambda}.e^{(-\lambda t)}\right]_0^\infty$$
$$\tau = 0 + \frac{1}{\lambda}.e^{(-\lambda 0)} = \frac{1}{\lambda}$$

- Mean/average lifetime: $\tau = \frac{1}{\lambda}$

Sample size after mean lifetime

- What is the sample size after 'mean life time' : $t = 1/\lambda$?

$$N(\tau) = N(0).exp(-\lambda\tau) = N(0).exp(-\lambda.\frac{1}{\lambda}) = \frac{N(0)}{e} = \frac{N(0)}{2.718}$$



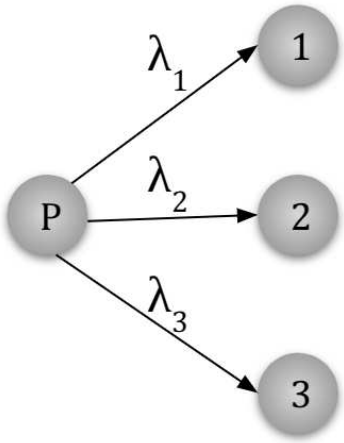
Half-Life

- Half-Life: Large sample of radioactive nuclei is reduced to half ($\frac{1}{2}$) of the number at t=0

$$N(\tau_{\frac{1}{2}}) = N(0).exp(-\lambda\tau_{\frac{1}{2}})$$
$$\frac{N(0)}{2} = N(0).exp(-\lambda\tau_{\frac{1}{2}})$$
$$exp(\lambda\tau_{\frac{1}{2}}) = 2$$
$$\lambda\tau_{\frac{1}{2}} = ln(2)$$
$$\tau_{\frac{1}{2}} = ln(2)/\lambda$$
$$\tau_{\frac{1}{2}} = \tau.ln(2)$$

Multiple Decay Modes

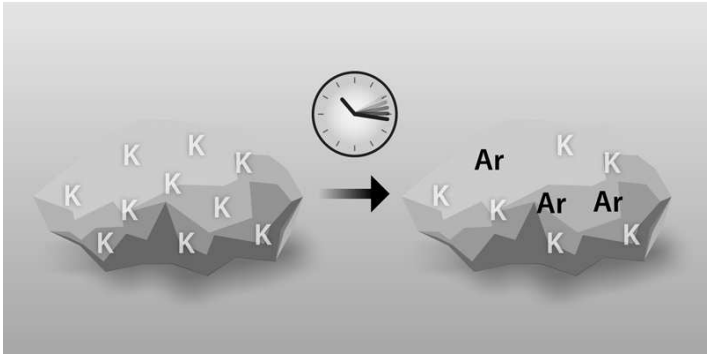
- Let us consider three competing branches in the decaying parent nuclei P to daughter nuclei 1, 2 and 3.
- The decay rates λ_1, λ_2 and λ_3 define probability per unit time for disintegration, P $\rightarrow$ 1, P $\rightarrow$ 2 and, P $\rightarrow$ 3, respectively.



- The total probability for disintegration (for P) is: $\lambda = \lambda_1 + \lambda_2 + \lambda_3$
- Relative probability for each branch decay, called the branching ratio, is the ratio of the respective decay rate to the total decay rate: $\mathcal{B}[\text{P} \rightarrow 1] = \lambda_1/\lambda$, $\mathcal{B}[\text{P} \rightarrow 2] = \lambda_2/\lambda$, $\mathcal{B}[\text{P} \rightarrow 3] = \lambda_3/\lambda$.
- And, also: $\frac{1}{\tau} = \frac{1}{\tau_1} + \frac{1}{\tau_2} + \frac{1}{\tau_3}$ (similar relation with half-life)

Radioactive Dating: $K - Ar$ Dating

- Parent Isotope: K^{40} is present in the molten rock (magma).
- Decay Process: A small fraction of the K^{40} in the rock decays $\tau_{1/2} = 1.25 \times 10^9$ y into Ar^{40}, a stable, inert noble gas.



Phys. Rev. Lett. 131, 052503 (2023)

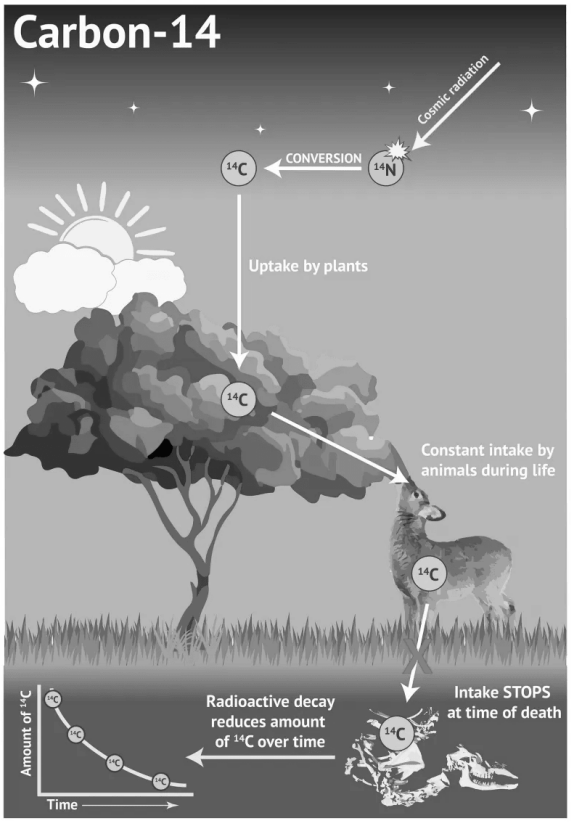
- When magma cools and solidifies, forming solid rock, the gaseous Ar^{40} escapes. This event effectively 'resets the clock.'
- From this point on, any produced Ar^{40}, by the decay of the trapped K^{40} is held within the rock's crystal structure.

$$t = \frac{\tau_{1/2}.ln(1+N_{Ar}/N_K)}{ln(2)}$$

- Dating rocks from Earth, the Moon, and meteorites shows a consistent maximum age of about 4.5 billion years.

Radioactive Dating: $C - 14$ Dating

- Radioactive C^{14} has a $\tau_{1/2}$ of 5,730 years.
- Cosmic rays bombard N^{14} in the upper atmosphere to produce C^{14} continuously.
- Clock starts when an organism dies, and exchange of stops C^{14}. The amount of C^{14} trapped within it begins to decay.
- By measuring the remaining ratio $C^{14} : C^{12}$, we can determine the time elapsed since the organism died.
- Ideal for dating organic materials up to $\sim 50,000$ years old.



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The End